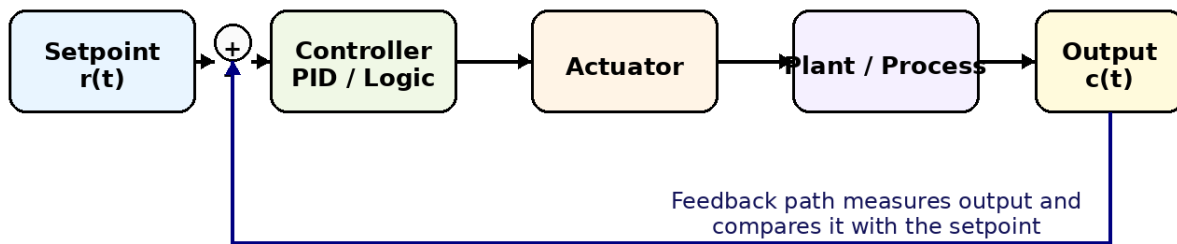


Industrial Automation & Control - Detailed Solutions Book

All assignment questions encountered so far / Weeks 2, 3, 6 and 7 / Separate Q&A volume

Basic Closed-Loop Control System

Goal: make output follow setpoint, reject disturbances, and remain stable.



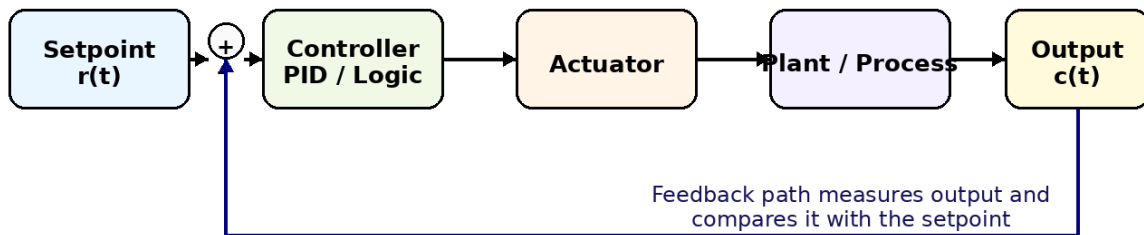
Prepared from the uploaded NPTEL assignment PDFs for Weeks 2, 3, 6 and 7.
Structured for exam revision with explanations from scratch.

What this volume contains

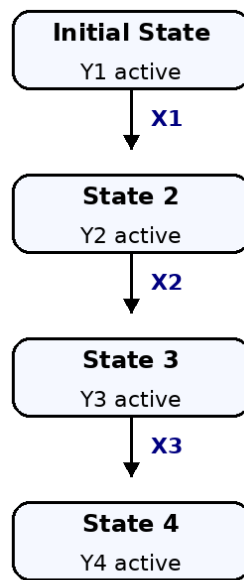
- Every question from the uploaded assignment PDFs encountered in this chat.
- Correct answer clearly marked.
- Reasoning written step by step in plain exam language.
- Extra mini-notes added wherever the question tests a core concept.

Basic Closed-Loop Control System

Goal: make output follow setpoint, reject disturbances, and remain stable.



From Description -> FSM / SFC -> Ladder Logic



Transition conditions -> assign outputs/actions to each state -> add timers/counters/interlock

Week 2 - Measurement Systems & DAS

Q1. In the general structure of a measurement system, Element IV corresponds to:

- a) Signal processing element
- b) Data presentation element**
- c) Sensing element
- d) Signal conditioning element

Correct answer:

Data presentation element

How to think about it

• In the common measurement chain, sensing comes first, then signal conditioning / processing, and the final stage presents the measured value to the user. Element IV is the final output/display stage, so it corresponds to the data presentation element.

Q2. Sensing is the _____ in a measurement system.

a) Signal processing element b) Data presentation element c) Display unit d) Process of converting physical quantity into another form	Correct answer: Process of converting physical quantity into another form
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How to think about it

- A sensor converts a physical variable such as temperature, pressure, or displacement into another form—usually an electrical signal—so that it can be processed. That conversion process is the essence of sensing.

Q3. In a weight measurement system, which of the following acts as the signal conditioning element?

a) Sensor b) Display c) Wheatstone bridge d) Power supply	Correct answer: Wheatstone bridge
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How to think about it

- The load cell is the sensor, but its tiny resistance change must be converted into a usable voltage. The Wheatstone bridge performs this conditioning/conversion step and therefore acts as the signal conditioning element.

Q4. Precision is the _____ of a measurement.

a) Repeatability b) Correctness c) Sensitivity d) Calibration	Correct answer: Repeatability
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How to think about it

- Precision tells you how closely repeated measurements cluster together under identical conditions. That is repeatability, not correctness. Correctness relates more to accuracy.

Q5. Which of the following devices does not involve energy storage elements and hence behaves as a zero-order system?

a) Thermometer b) Accelerometer c) Seismometer d) Potentiometer	Correct answer: Potentiometer
---	--

How to think about it

- A zero-order instrument responds instantaneously and has no energy storage element. A potentiometer directly maps displacement to voltage with negligible storage. Thermometers, accelerometers, and seismometers have thermal or mechanical inertia.

Q6. A second-order system having real and unequal roots is classified as:

a) Underdamped b) Overdamped c) Critically damped d) Undamped	Correct answer: Overdamped
---	---

How to think about it

- For a standard second-order system, real and unequal roots occur when damping ratio $\zeta > 1$. That is the overdamped case. Underdamped systems have complex roots; critical damping has repeated real roots.

Q7. A 4-bit R-2R ladder DAC has reference voltage $V_{ref} = -5\text{ V}$. What is the analog output for digital input 1110?

a) 3.75 V	Correct answer: 4.375 V
b) 1.875 V	
c) 4.375 V	
d) 2.1875 V	

How to think about it

- For a 4-bit DAC, 1110 corresponds to decimal 14. Magnitude of output = $(14/16) * |V_{ref}| = (14/16) * 5 = 4.375\text{ V}$. The ladder plus op-amp configuration shown makes the final listed positive output 4.375 V.
- For an n-bit DAC: output step (LSB) = full-scale range / $(2^n - 1)$ or approximately $V_{ref}/2^n$ depending on the convention.
- For an R-2R ladder, convert binary value to decimal and scale it by the reference.

Q8. An 8-bit DAC with digital input changing from 00000000 to 00001001 has 0 to 10 V output range. Find the analog output change.

a) 0.353 V	Correct answer: 0.353 V
b) 0.256 V	
c) 39.2 mV	
d) 18.5 mV	

How to think about it

- LSB size = $10/255 \approx 0.0392\text{ V}$ for an 8-bit DAC covering 0–10 V. Change from 0 to 9 counts gives $9 \times 0.0392 \approx 0.353\text{ V}$.
- For an n-bit DAC: output step (LSB) = full-scale range / $(2^n - 1)$ or approximately $V_{ref}/2^n$ depending on the convention.

- For an R-2R ladder, convert binary value to decimal and scale it by the reference.

Q9. An 8-bit successive-approximation ADC is driven by a 5 MHz clock. Assuming reset condition, find the conversion time.

a) 4.4 μ s b) 2.8 μ s c) 3.2 μ s d) 1.6 μs	Correct answer: 1.6 μs
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How to think about it

- A SAR ADC typically needs about n clock periods for n bits. Here n = 8. Clock period = $1/(5 \text{ MHz}) = 0.2 \mu\text{s}$. So conversion time $\approx 8 \times 0.2 = 1.6 \mu\text{s}$.
- A SAR ADC typically requires about n clock cycles for n bits.
- Clock period $T = 1/f$. Conversion time $\approx nT$.

Q10. Choose the minimum sampling frequency so that $x(t) = 5 + 2\sin(2\pi \cdot 50 t) + 5\cos(2\pi \cdot 150 t + \pi/3)$ can theoretically be reconstructed.

a) 150 Hz b) 250 Hz c) 450 Hz d) 350 Hz	Correct answer: 350 Hz
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How to think about it

- The highest frequency component is 150 Hz. Nyquist requires $f_s > 2f_{\text{max}} = 300 \text{ Hz}$. Among the options, 350 Hz is the minimum frequency above 300 Hz, so it is correct.
- Step 1: identify highest frequency component. Step 2: apply $f_s > 2 f_{\text{max}}$. Step 3: choose the smallest option above that value.

Q11. A 0-100°C sensor has sensitivity 4 mV/°C. An 8-bit ADC with 10 V reference is used. Find integer amplifier gain for maximum possible resolution.

a) 42	Correct answer: 25
b) 25	
c) 33	
d) 16	

How to think about it

- Sensor full-scale output = $100 \times 4 \text{ mV} = 0.4 \text{ V}$. To utilize the ADC range well, amplify 0.4 V close to 10 V: gain $\approx 10/0.4 = 25$. So the best integer gain is 25.

Q12. Choose the correct option: I) Flash ADC is suitable for more than 8 output bits. II) SAR ADC uses a trial-and-error successive approximation register. III) Typical resistance values in an R-2R ladder can vary from 2.5 kΩ to 10 kΩ. IV) Anti-aliasing filter is a low-pass filter.

a) I and II are correct	Correct answer: II, III and IV are correct
b) II and III are correct	
c) II, III and IV are correct	
d) III and IV are correct	

How to think about it

- I is false because flash ADC becomes impractical at high bit counts due to huge comparator count ($2^n - 1$). II is true. III is acceptable because only the ratio R:2R matters and practical values may indeed span such values. IV is true because anti-aliasing uses a low-pass filter before sampling.
- For an n-bit DAC: output step (LSB) = full-scale range / ($2^n - 1$) or approximately $V_{\text{ref}}/2^n$ depending on the convention.

- For an R-2R ladder, convert binary value to decimal and scale it by the reference.

Week 3 - Automatic Control & PID

Q1. Find the steady-state error for unit step input to the unity-feedback system with forward path $G(s)=120(s+2)/((s+3)(s+4))$.

a) 0.024	Correct answer: 0.048
b) 0.032	
c) 0.048	
d) 0.083	

How to think about it

- For a unity-feedback type-0 system with step input, $e_{ss} = 1/(1+K_p)$, where $K_p = \lim_{s \rightarrow 0} G(s)$. Here $K_p = 120(2)/(3 \cdot 4) = 240/12 = 20$. Hence $e_{ss} = 1/(1+20) = 1/21 \approx 0.0476 \approx 0.048$.
- For unity-feedback type-0 system and step input: $e_{ss} = 1 / (1 + K_p)$.
- Compute K_p by substituting $s = 0$ in $G(s)$.

Q2. Write the output response $C(s)$ in terms of reference input $R(s)$ and disturbance $D(s)$ for the block diagram in Fig. 2.

a) a	Correct answer: Option b
b) b	
c) c	
d) d	

How to think about it

- The disturbance enters between $G_1(s)$ and $G_2(s)$. With negative feedback $H(s)$, the closed-loop denominator is $1+G_1(s)G_2(s)H(s)$. The numerator contribution becomes $G_2(s)[G_1(s)R(s)+D(s)]$. Therefore $C(s) = G_2(s)/(1+G_1G_2H) \cdot [G_1R + D]$, which matches option b.

Q3. Match controller structures: one degree of freedom, two degree of freedom, three degree of freedom with the given diagrams.

a) A-I, B-II, C-III b) A-III, B-I, C-II c) A-III, B-II, C-I d) A-II, B-I, C-III	Correct answer: A-III, B-I, C-II
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How to think about it

- A 1-DOF controller has only the basic feedback path, so it matches the simplest diagram. A 2-DOF structure adds a separate reference path weighting/feedforward path. A 3-DOF structure additionally handles disturbance/noise shaping independently. That mapping corresponds to option b.

Q4. Possible reasons of phase lag inside a feedback loop are:

a) Delay between sensor input and plant output b) Delay between controller output and plant input c) Time constants in sensor, plant and actuator d) All of the above	Correct answer: All of the above
---	---

How to think about it

- Phase lag can come from transport delay, actuator/sensor delay, and dynamic elements with time constants. Every element that stores energy or delays signal transmission contributes lag, so all listed causes are valid.

Q5. The gain K of the tachogenerator shown mainly influences:

a) Phase shift at very high frequencies b) Phase shift at very low frequencies c) Natural frequency of oscillation	Correct answer: Peak overshoot
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d) Peak overshoot	
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How to think about it

- Tacho-generator feedback provides derivative-like velocity feedback. Its most visible effect is increased damping, which reduces overshoot in the transient response. It does not primarily determine low- or high-frequency phase shift in the way the options suggest.

Q6. If the actuator output is in saturation region, the system behaves like _____ control.

a) Open loop b) Closed loop c) Linear d) Feedforward	Correct answer: Open loop
---	--

How to think about it

- When an actuator saturates, further controller effort cannot produce additional plant input. The feedback loop effectively loses control authority and behaves as though open-loop over that region.

Q7. Which statement about a PID controller equation is true?

a) The integral term is multiplied by $K_p T_i$ b) The derivative term is independent of proportional gain c) The integral term coefficient is K_p/T_i d) The proportional term depends on derivative time	Correct answer: The integral term coefficient is K_p/T_i
---	--

How to think about it

- In the standard ideal/parallel PID form, $u(t) = K_p[e(t) + (1/T_i) \int e \, dt + T_d \, de/dt]$. Therefore the coefficient multiplying the integral part is K_p/T_i .
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q8. What is the relationship between proportional band (PB) and proportional gain (Kp)?

a) PB directly proportional to K_p b) PB inversely proportional to K_p c) PB independent of K_p d) PB equal to K_p	Correct answer: PB inversely proportional to K_p
---	--

How to think about it

- Proportional band is the input percentage change needed to drive the output through full scale. A larger K_p means a smaller band is needed, so $PB \propto 1/K_p$.
- Remember: larger gain -> smaller proportional band.
- So PB is inversely proportional to K_p .

Q9. Derivative time is defined as the time required for the _____ action to equal the _____ action for a ramp error signal.

a) Integral, Proportional b) Proportional, Integral c) Derivative, Integral d) Proportional, Derivative	Correct answer: Proportional, Derivative
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How to think about it

- For a ramp error, derivative action produces a constant output proportional to slope, while proportional output grows linearly with time. Derivative time T_d is defined as the time at which proportional action catches up to the derivative action. So it is the time required for proportional action to equal derivative action.

Q10. Assertion: Derivative block is normally applied to incremental signal Δu rather than total signal u . Reason: This prevents sudden jumps in plant output during auto-manual mode transfer.

a) Both true and reason explains assertion	Correct answer: Both true and reason explains assertion
b) Both true but reason does not explain	
c) Assertion true, reason false	
d) Assertion false, reason true	

How to think about it

- Applying derivative to abrupt total-output changes causes derivative kick. Using the incremental signal helps avoid sudden jumps during transfer, so both statements are true and the reason correctly explains the assertion.

Q11. Assertion: The derivative term in a PID controller is usually limited. Reason: High-frequency noise can cause large rapid changes in the derivative output.

a) A true, R false	Correct answer: Both true and R explains
b) A false, R true	
c) Both true but R does not explain	
d) Both true and R explains	

How to think about it

- Derivative amplifies rapid signal changes, and measurement noise typically contains high-frequency content. Therefore practical PID controllers limit or filter derivative action. The reason directly explains the assertion.

Q12. The incremental realization of a PID controller, where output is calculated as the change in control signal, is also known as the ____ form.

a) Position form	Correct answer: Velocity form
b) Velocity form	
c) Acceleration form	
d) Continuous form	

How to think about it

- When the controller computes Δu rather than absolute u each scan, it is called the velocity or incremental form.

Position form computes the full output directly.

Week 6 - PLC, Relay Logic & Scan Cycle

Q1. For the domestic central heating system of Fig. 1, choose the correct combination of sensors that turns on boiler output Q2.0.

a) a	Correct answer: Option d
b) b	
c) c	
d) d	

How to think about it

- The boiler should run only when the clock enables operation and at least one heating demand exists. In the given diagram, option d matches the required enabling logic from the clock and temperature-demand sensors as shown in the ladder network. The key idea is: boiler ON = allowed time AND (room needs heat OR tank needs heat).

Q2. For the relay ladder diagram in Fig. 2, write the combination of input contacts so that output coil OP1 energizes.

a) a b) b c) c d) d	Correct answer: Option a
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How to think about it

- Read ladder logic left-to-right: series contacts represent AND, parallel branches represent OR, and normally-closed contacts introduce negation. The arrangement in Fig. 2 resolves into $OP1 = ((IN1 + IN4 + \neg IN6)(\neg IN2 + IN5)(IN3 + IN7))$, which is option a.

Q3. For the relay ladder diagram in Fig. 3, choose the correct logical expression of input contacts that turns on motor M1 (Q2.0).

a) $(I0.0)(I0.1)(I0.2)$ b) $(I0.0)(I0.1)$ c) $(I0.0)+(I0.1)$ d) $(I0.0)(I0.1)+(I0.2)$	Correct answer: $(I0.0)(I0.1)$
--	---

How to think about it

- In the rung for M1, the motor energizes only when the required series path is complete. Parallel or branch elements in the second rung affect the other output, not M1. Therefore M1 depends on the series condition I0.0 AND I0.1, matching option b.

Q4. The relay ladder logic shown in Fig. 4 resembles which flip-flop?

a) T b) D	Correct answer:
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c) RS	RS
d) JK	

How to think about it

- The rung shows a self-holding/latching structure with a start path and a break/reset path using the output contact itself. That behavior is characteristic of set-reset (RS) action, not toggle (T), data (D), or JK behavior.

Q5. In the relay ladder diagram of Fig. 5, coil PO1 will energize if:

a) IN1 closed, IN4 closed, and IN5 open	Correct answer:
b) IN7 open, IN2 closed, and IN3 open	IN1 closed, IN4 closed, and IN5 closed
c) IN7 closed and IN6 open	
d) IN1 closed, IN4 closed, and IN5 closed	

How to think about it

- Trace the current path to the output. The valid complete conduction path occurs through the lower branch when IN1 and IN4 are closed, together with the branch condition requiring IN5 closed. The other listed combinations do not complete the rung as shown.

Q6. Choose the correct ladder logic for the scenario: lamp Y430 ON if pump X400 is running and pressure sensor X401 reading is satisfactory, or if lamp-test switch X402 is closed.

a) a	Correct answer:
b) b	Option c
c) c	
d) d	

How to think about it

- The specification is $Y430 = (X400 \text{ AND } X401) \text{ OR } X402$. In ladder logic, AND is series and OR is parallel. So the correct diagram must have one branch with X400 in series with X401, in parallel with another branch containing X402 alone. That is option c.

Q7. For ladder diagram in Fig. 6 using JSR, SBR, RET, choose the correct option about statements i) and ii).

a) Both true b) i true ii false c) i false ii true d) Both false	Correct answer: i false, ii true
---	---

How to think about it

- JSR transfers execution to the subroutine. Logic after SBR belongs to the subroutine, not the main program, so statement i is false. When input In2 occurs in the active context, Out1 occurs as shown, so statement ii is true.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q8. For ladder diagram in Fig. 7 using JSR, SBR, RET, choose the correct option about statements i) and ii).

a) Both true b) i true ii false c) i false ii true d) Both false	Correct answer: Both true
--	--

How to think about it

- Input In1 causes execution to jump into the subroutine sequence, where Out2 is produced as shown. After Out3 and RET, the program resumes main flow and waits for the next qualifying input such as In2. Hence both statements are true.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q9. Choose the correct input-output waveform for the ladder logic in Fig. 8.

a) a	Correct answer: Option c
b) b	
c) c	
d) d	

How to think about it

- The rung has one branch with A AND B, and another branch with NOT A AND NOT B. Therefore output = $AB + \bar{A}\bar{B}$, which is XNOR behavior: output is high when A and B are equal, and low when they differ. The waveform matching that is option c.

Q10. Which Sequential Function Chart correctly describes the ladder diagram in Fig. 9?

a) a	Correct answer: Option b
b) b	
c) c	
d) d	

How to think about it

- The ladder logic has Y1 driven directly by X1 in one path, while Y2 depends on X2 and X3 in another sequence branch. The SFC that captures that progression and dependency pattern is option b.

Q11. For the SFC in Fig. 10, output Y2 will occur if output Y1 has been realized and: i) both X2 and X3 have been realized, ii) only X2 has been realized.

a) Both true b) i false ii true c) i true ii false d) Both false	Correct answer: Both true
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How to think about it

- From State 1, the chart branches to State 2 on X2 and to State 3 on X3. Y2 belongs to State 2, so Y2 occurs if X2 is realized regardless of whether X3 also occurs. Thus both statements are true.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q12. For the SFC in Fig. 11, output Y4 will occur if output Y2 has been realized and: i) output Y3 has occurred, ii) condition X4 has been realized.

a) Both true b) i false ii true c) i true ii false d) Both false	Correct answer: i false, ii true
--	--

How to think about it

- Y4 belongs to the transition path from the Y2 branch via condition X4. The occurrence of Y3 is not required for Y4. Therefore statement i is false and statement ii is true.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Week 7 - Timers, Counters, RLL Syntax & Structured Design

Q1. Match timer forms with their input-output plots: A) On-delay timer, B) Off-delay timer, C) Pulse timer.

a) A-I, B-II, C-III b) A-III, B-I, C-II c) A-III, B-II, C-I d) A-II, B-III, C-I	Correct answer: A-II, B-III, C-I
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How to think about it

- An on-delay timer waits before turning output ON after input goes high, so it matches the plot where the output starts late. An off-delay timer keeps output ON briefly after input goes low, matching the extended tail plot. A pulse timer gives a fixed-width pulse when triggered, matching the short pulse output. That mapping is A-II, B-III, C-I.

Q2. Considering input-active and TON timing in Fig. 1, choose the correct statement about the lamp.

a) Lamp is only on when Timer 0 is off b) Lamp is only on when Timer 1 is on c) Lamp is only on when Timer 0 is on d) Lamp is only on when Timer 0 and Timer 1 are on	Correct answer: Lamp is only on when Timer 0 is on
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How to think about it

- The lamp rung is driven directly by the Timer 0 contact. Therefore the lamp follows the done/energized status of Timer 0. Timer 1 affects the upper logic but not the lamp directly in the shown rung.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q3. A traffic-light sequence is red → red+amber → green → amber → repeat. An SFC is given. Choose the correct ladder diagram.

a) a b) b c) c d) d	Correct answer: Option a
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How to think about it

- The correct ladder must implement the states in that exact order with the listed dwell times: red (2 min), red+amber (20 s), green (2 min), amber (20 s). Option a is the diagram that preserves that state sequence and timer placement correctly.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q4. Fig. 3 shows a ladder diagram involving counter C460, inputs X400 and X401, internal relays M100 and M101, and output Y430. Assuming M100, M101 and Y430 are de-energized initially, decide whether: i) Y430

switches on at the eighth pulse to X400. ii) Y430 switches off at the start of the ninth pulse to X400.

a) Both i and ii are true b) i false, ii true c) i true, ii false d) Both false	Correct answer: Both i and ii are true
--	--

How to think about it

- The counter preset is 8, so its done condition becomes true at the eighth qualifying pulse. That causes the output logic to energize Y430. Once the next pulse begins, the latch/reset condition changes as per the rung logic and Y430 turns off at the start of the ninth pulse. Hence both statements are true.

Q5. For the ladder shown in Fig. 4, there is an output from OP02 every time: i) input IN01 has closed 5 times. ii) input IN02 has closed 5 times.

a) i true, ii false b) i false, ii true c) Both true d) Both false	Correct answer: i false, ii true
---	--

How to think about it

- The counter is wired to the IN02 branch, not IN01. OP01 is the counter-done output, and OP02 follows OP01 in the second rung. Therefore OP02 occurs when IN02 has been counted 5 times, not IN01.

Q6. Choose the correct ladder logic for a conveyor that counts 10 bottles using photocell IN02, started by push button IN01, then activates solenoid OP01 and resets the counter.

a) a b) b	Correct answer:
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c) c	Option c
d) d	

How to think about it

- The correct design needs: start command, counting of bottle pulses from IN02, counter preset 10, output OP01 when count completes, and reset path coordinated with OP01. Option c is the one that correctly integrates start, count, done action, and reset behavior.

Q7. Why is a very systematic design process emphasized for industrial control applications?

a) Because it reduces programming time b) Because industrial control programs are simple c) Because it eliminates testing d) Because programming errors can be very expensive or even cost human lives	Correct answer: Because programming errors can be very expensive or even cost human lives
--	--

How to think about it

- Industrial control mistakes can damage equipment, waste production, create unsafe motions, or injure people. That is why systematic design, review, simulation, and validation are essential.

Q8. Automatic generation of RLL programs will be error-free only if:

a) PLC scan time is small b) Sensors never fail c) Formal specifications and models are correct d) Manual control is not provided	Correct answer: Formal specifications and models are correct
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How to think about it

- Automatic code generation can only be as good as the model/specification it starts from. If the model is wrong or incomplete, the generated ladder logic will also be wrong. So correctness depends fundamentally on correct formal specifications.

Q9. Directly programming outputs from inputs becomes unreliable in large systems primarily because:

a) PLCs cannot handle large programs	Correct answer: Systems exhibit memory and depend on current states
b) Systems exhibit memory and depend on current states	
c) Inputs change too frequently	
d) Sensors generate conflicting signals	

How to think about it

- Large sequential systems are state-dependent. The same input may require different outputs depending on what has already happened. Pure input-output mapping ignores memory and history, so it becomes unreliable. That is why states, FSMs, and SFCs are needed.

Q10. An initial step in a Sequential Function Chart (SFC) is executed:

a) After power-on and after program reset	Correct answer: After power-on and after program reset
b) Only once after power-on	
c) Only during fault recovery	
d) Only if manually triggered	

How to think about it

- The initial step is the defined starting state of the sequence. Whenever the controller powers up or the sequence is reset, execution resumes from that initial step.

- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q11. If two alternative transitions become true simultaneously in an SFC, what happens?

a) Both branches execute concurrently b) Execution stops due to conflict c) PLC chooses randomly d) The leftmost transition is selected	Correct answer: The leftmost transition is selected
---	--

How to think about it

- For alternative branches, only one path may be taken. Standard SFC semantics resolve simultaneous true alternatives by priority, typically left-to-right, so the leftmost enabled transition is selected.
- Translate the chart carefully: active step -> associated output active. Transition condition true -> move to next step.
- For alternative branches, use priority. For parallel branches, require synchronization.

Q12. Which of the following is illegal in Sequential Function Charts?

a) Nested parallel branches b) Jumping into or out of a parallel sequence c) Jumping after a transition d) Program termination using dead states	Correct answer: Jumping into or out of a parallel sequence
--	---

How to think about it

- Parallel sequences have strict synchronization rules. Arbitrarily jumping into or out of a parallel branch breaks activation logic and synchronization, so that is illegal in SFC design.

Fast solving strategy for exams

- Measurement questions: identify the stage of the chain first—sensor, conditioning, processing, display.
- Control questions: separate steady-state ideas from transient ideas.
- PID questions: map each term to its job—P speed, I zero offset, D damping.
- PLC questions: redraw series/parallel logic as Boolean expression before choosing the option.
- SFC questions: ask which state is active, then which transition fires next.